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Analyzing Greenland's Geochemistry through Machine Learning

Research Project Report

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In progress

Abstract

TO DO

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1 Introduction

Greenland is one of the world's most mineral-rich but least-explored regions. Over decades, the Geological Survey of Denmark and Greenland (GEUS) [1] has assembled an extensive public database of geochemistry samples, exploration reports, mineral occurrences, intrusions, geological maps, and license boundaries, all accessible through GEUS web services. This data is heterogeneous in coverage, quality, and completeness, which makes it an interesting and realistic setting for applying machine learning. The aim of this project is to use machine learning to analyse geochemical data from GEUS and explore what can be learned from it. For example, identifying patterns across elements, reconstructing missing measurements, mapping spatial trends, or relating geochemistry to known mineral occurrences.

1.1 Research questions

The project was organized around two major problems:

RQ1 — missing geochemistry. Given a sample predict an held-out element from the sample's other elements and its spatial neighborhood.

RQ2 — unexplored regions. Predict concentrations in an held out geological area rather than at random, avoiding spatial leakage.

This report focuses on RQ1, while setting some ground for the RQ2.

1.2 Contributions

This project applies one identical pipeline to two GEUS datasets with different sampling geometries. I contribute with the following:

Contrasting different KNN integrations. Comparing distinct methods of encoding a spatial neighborhood as tabular data, across seven neighborhood sizes.

Evidence that spatial context is largely redundant. A strong baseline obtained only by a sample's own geochemistry doesn't gain much with geophysical or spatial information. Some elements are exceptions, however.

Comparison between hand-built vs learned neighbor aggregation. Two different graph neural networks (GNN), with distance-aware and unaware edges, are evaluated against the handbuilt KNN encodings.

2 Data and Methods

2.1 The datasets

Two public geochemical datasets, that differ sharply in sampling geometry (Figure 1 and table 1), were picked and pre-processed by the exact same code. Whole rock is an exploration sampling within a narrow band of South-West Greenland, whilst Stream Sediment covers the entire margin of Greenland.

Table 1: The two datasets, before and after preprocessing.

	Whole rock	Stream sediment
Raw rows	26 068	20 158
Raw element columns	76	523
Merged rows	6 957	17 463
Merged element columns	76	93
Target elements (coverage $\geq$ 1 000)	48	71
(After pre-processing)		
Unique coordinate pairs	4 498	16 542
Co-located samples	3 458 (49.7 %)	1 759 (10.1 %)
Max. samples at one location	67	11
Longitude span	11.7°	59.5°
Latitude span	10.3°	23.8°

Each source file is accompanied by a JSON meta file naming the sample number identifier, metadata, coordinates, and geophysical columns. The most important, geochemical columns, are whatever remains from this,

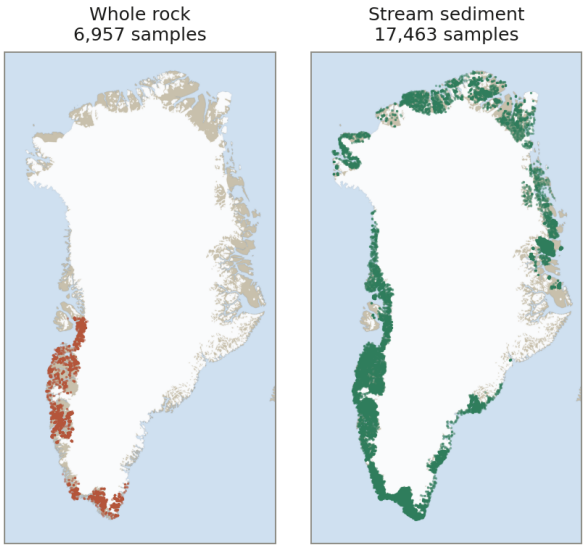


Figure 1: Sampling locations post-pre-processing on both datasets

2.2 Preprocessing

The preprocessing of data consists on the following six steps:

1. **Mask below-detection values.** non-NaN entries that present negative values are usually interpreted as below-detection limit (BDL). Rather than keeping these values as is, or trying to perform educated approaches, BDL values are treated as NaN.

2. **Merge analytical columns.** Some columns represent a measurement of the same element but across different laboratories and methods. These different methods differ in detection limits, biases, instruments, etc, even if measured on the same sample for the same element, so they are aggregated by the median as a neutral design choice. This is what makes the Stream Sediment dataset move from 523 columns to 93.

3. **Merge by sample number.** Similarly to the step above, some rows are repeated across the same extracted sample, but measured by different methods. The aggregation is done in the exact same way: median across the rows depicting the same sample number.

4. **Log-transform.** After merging both dimensions, the numerical data is log_{10} transformed to follow the natural distribution of geochemistry. An added advantage is that this process diminishes the negative effect of outliers and sentinels.

5. **Coverage filter.** Element columns measured in fewer than 10 samples are completely dropped from the dataset, as columns with mostly NaN wouldn't add meaningful information for inference.

6. **Splits.** All experiments are done on the very same five-fold cross-validation. As we are still in the context of RQ1, there is no spatial holdout, and the samples are randomized, leading to a perfect 80/20 split in every fold (Figure 2).

2.3 Targets and evaluation

Every element measured in at least 1 000 samples is a target. Leading to 48 and 71 total targets on Whole Rock and Stream Sediment, respectively.

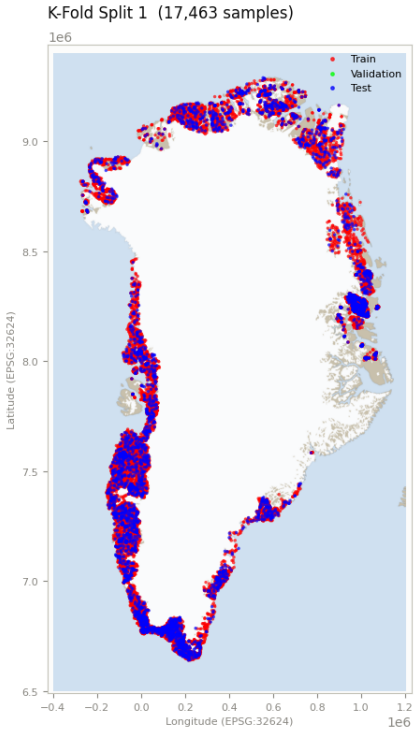


Figure 2: Example of sample distribution across splits in 5-Fold CV on Stream Sediment dataset.

Each target is predicted from the sample's remaining elements and scored by R^2 over the five random folds. Samples that did not measure the target element are naturally discarded.

Once again, since we are in the context of RQ1, spatial leakage can be a problem due to lack of spatial CV. That is, a model trained on a certain sample can be guessing a sample just a few meters away, which can be considered duplicates in certain perspectives.

2.4 KNN Encoding

Neighbor features are built at training time, per fold. This means that a k-d tree is fitted on the training samples only, so a test sample never recognizes another test sample as a neighbor, which is a very straightforward method to block data leaks. Additionally, self-matches and co-located neighbors are discarded, meaning that a neighbor is necessarily at a different geographical location, not just a different sample from the same exploration site.

Four different types of encodings are compared. With k neighbours and m elements they differ in how much they aggregate (Table 2).

Table 2: The four neighbour encodings. Column counts are for $k = 20$, $m = 87$ (stream sediment).

Variant	Content	Added columns
A	Mean of the target element over k neighbours	1
B	The k individual target values	20
C	Mean of every element over k neighbours	87
D	Every element of *every* neighbour	1 740

2.5 Geophysical features

Geochemical neighbors only exist where somebody has already explored. Geophysical grids, on the other hand, are measured everywhere. Therefore, on unexplored ground, they are a motivated input.

Three geophysical attributes were added to each entry of both datasets: Magnetic anomaly, geothermal heat flow, and depth to Moho.

Despite wanting to introduce more geophysical features, I wasn't able to find more grids that properly covered all points in our dataset on the GEUS platform.

2.6 Models

Gradient-boosted decision trees remain one of the strongest machine learning methods on tabular data with this size and heterogeneity. The main model is the histogram-based gradient boosting (HGB) implementation in scikit-learn. As a comparison, a simple random forest (RF) is also run on the same pipeline, as to check that conclusions are not an artifact of model bias.

2.7 Graph neural network

A GNN offers a different way to use the sample neighborhood. Rather than aggregating by mean, or enumerating individual values into fixed columns, the relationships and their relevance is learned automatically. The graph presents one node per sample, and directed edges to the respective k nearest neighbors. So each sample sees the exact same neighborhood in the GNN as it saw in the hand-built version. Each edge carries $\log(1 + d)$, with d in kilometers. The node features are the sample's own elements. Since the network cannot consume NaN values, each element has two channels: A value channel where it is zero whenever the original value is NaN; And a presence/absence channel to serve as a flag.

Two different architectures are compared: A GraphSAGE, which is distance-blind by construction, and a graph attention network (GAT), that computes attention weights per edge from the distance, wich is something the previous KNN variants were missing. Other than this structural distinction, both use two message-passing layers of width 64, a skip connection from the node's features to a two layer head, dropout 0.1, and Adam optimizer at learning rate 10^{-2} with 10^{-4} weight decay.

GNN experiments were run only on Stream Sediment as it was the only dataset where standalone spatial signal was strong enough to justify a learnt aggregation.

3 Results

3.1 Own chemistry sets a high baseline

Predicting a held-out element from a sample's remaining elements reaches a mean R^2 of 0.7983 on whole rock (48 targets) and 0.8676 on stream sediment (71 targets). This is the number every spatial method below must improve upon, and it is high because elements are strongly co-determined: a sample rich in SiO_2 is necessarily poor in something else.

The two figures are not directly comparable to one another, since the target sets differ — whole rock's 48 elements are proportionally more trace metals, stream sediment's 71 include a large suite of near-collinear rare earths that are easy to predict. Every comparison in this chapter is therefore made *within* a dataset.

Among the three tabular learners, gradient boosting was both the most accurate — on every one of the ten whole-rock elements tested — and the fastest, at mean R^2 0.8388 in 155 s against 0.8207 in 388 s for a 200-tree random forest and 0.8215 in 199 s for XGBoost. It is used throughout. XGBoost's tie with the random forest is consistent with its competitive reputation resting on

extensive tuning that its defaults (`max_depth` = 6, `learning_rate` = 0.3) do not supply for a dataset of this size and correlation structure.

3.2 A neighbourhood should be pooled, not enumerated

Table 3 gives the four encodings at $k = 20$. The shape is the same on both datasets: variant C is best, variant D lands at or below the baseline, and every gain is smaller than one fold standard deviation — 0.82σ on whole rock, 0.91σ on stream sediment.

Table 3: The four neighbour encodings, $k = 20$, co-located neighbours excluded. Fold standard deviation is 0.0180 (whole rock) and 0.0137 (stream sediment).

	Whole rock (48)		Stream sediment (71)	
Variant	Mean R^2	Δ	Mean R^2	Δ
Baseline (no neighbours)	0.7983	—	0.8676	—
A	0.8043	+0.0059	0.8757	+0.0081
B	0.8050	+0.0066	0.8724	+0.0048
C	0.8132	**+0.0149**	0.8801	**+0.0125**
D	0.7982	−0.0001	0.8659	−0.0017

Sweeping k from 1 to 100 (Figure 3) separates aggregation from dimensionality, and two trends run in opposite directions on both datasets. Variant C improves with k and plateaus around $k \approx 20$–50: because it averages over the neighbourhood, its width stays at m columns whatever k is, so more neighbours only smooth sampling noise and cost nothing. Variant D degrades monotonically, and the decline tracks its column count exactly — 1 740 columns at $k = 20$, 8 700 at $k = 100$. D contains strictly more information than C and uses it strictly worse.

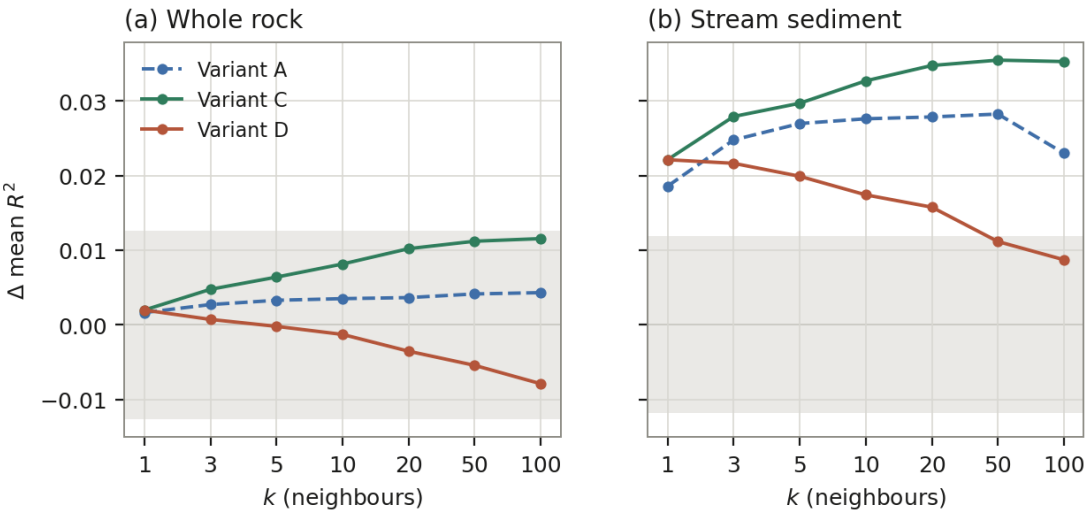


Figure 3: Change in mean R^2 over the no-neighbour baseline as neighbourhood size k varies, on the ten best-covered elements of each dataset. The grey band is one fold standard deviation. Variant C (pooled) rises and plateaus; variant D (enumerated) falls in step with its column count. Absolute levels reflect the ten-element target sets (Section 3.8); the shape is what this figure establishes.

This is the most robust result in the report — it holds across two datasets, two model families and seven values of k — and it fixes $k = 20$ for everything that follows: the onset of the plateau, and effectively free for variant C, whose column count does not depend on k.

3.3 Where the gain lands

A pooled mean over dozens of elements averages together very different behaviours. Binning targets by *their own baseline score* separates the elements own chemistry already predicts from those it cannot (Figure 4).

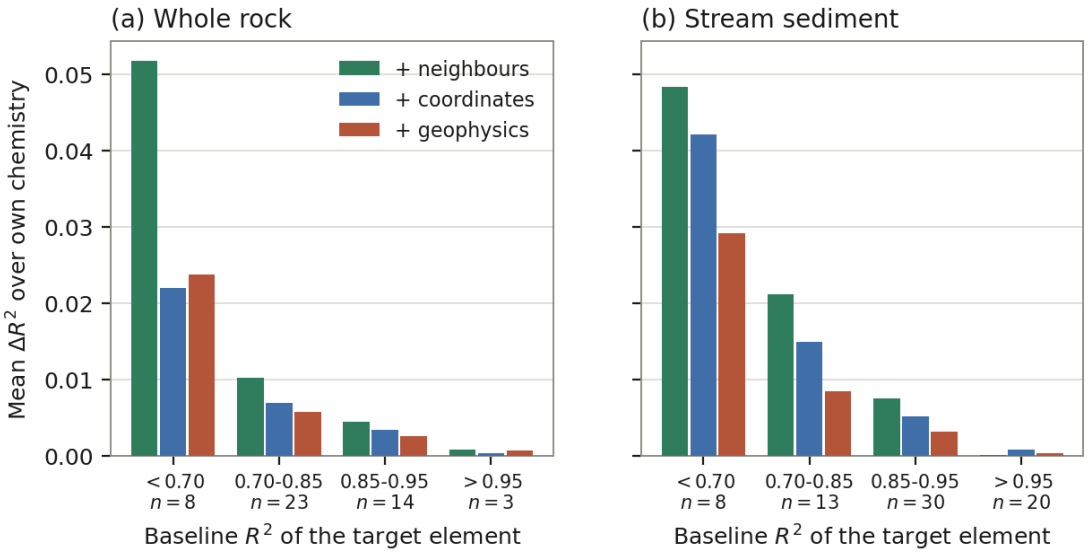


Figure 4: Mean improvement over own chemistry, with targets grouped by how well own chemistry already predicts them. Two datasets with different sampling geometries, different target sets and different element counts produce almost the same curve.

The decay is monotone on both datasets and the two curves are nearly identical, despite the datasets differing in sample count, spatial extent, co-location rate and target composition. Neighbours are worth +0.0517 to the eight whole-rock targets below 0.70 (mean baseline 0.5920) and +0.0484 to the eight stream-sediment ones (0.5846); above 0.95 they are worth +0.0008 across three whole-rock targets and +0.0001 across twenty stream ones — unsurprisingly, since there is no headroom left to fill. **This, rather than any pooled mean, is the central result of RQ1**, and it explains mechanically why a pooled mean is fragile: changing the target set changes the mixture of difficulty bands, and therefore the average, without anything about the underlying behaviour changing at all.

3.4 Which elements benefit

The elements gaining most are not an arbitrary assortment, and they recur across the two datasets (Table 4). Arsenic, antimony, lead and caesium appear in both top-six lists, computed on different samples, different target sets and different sampling geometries.

These are, bromine aside, the classical hydrothermal pathfinder suite together with mobile alkalis: elements whose distribution is governed by regional-scale processes such as mineralisation and alteration rather than by bulk rock composition, and which therefore have something for a neighbourhood to say that the host rock's major elements do not. Uranium is the extreme case, rising on stream sediment from 0.594 on own chemistry alone to 0.814 with neighbours. It is also the only element with a reliable geophysical correlation in this data (+0.28 with magnetic anomaly, −0.34 with Moho depth, $n = 3\,227$), which is coherent — uranium concentrates in evolved felsic crust, which is thick and weakly magnetic.

Table 4: The six elements with the largest gain from neighbour features on each dataset, $k = 20$. Four elements (As, Sb, Pb, Cs) appear on both lists.

Whole rock		Stream sediment	
Element	$\Delta\ R^2$	Element	$\Delta\ R^2$
As	+0.096	U	**+0.220**
Au	+0.092	As	+0.060
Sb	+0.070	Br	+0.043
Cs	+0.048	Sb	+0.041
S	+0.040	Pb	+0.034
Pb	+0.027	Cs	+0.031

3.5 Chemical isolation is the common cause

Sections 3.3 and 3.4 give two descriptions of the same set of elements — ones own chemistry predicts poorly, and ones that are geochemically mobile — which invites the question of what the two have in common. Measuring the chemistry directly answers it. For each element, take the single strongest Spearman correlation it has with any other element in the same sample, computed pairwise-complete over the raw log concentrations and independently of any model. Call that its *companion strength*: how good a proxy the sample already carries.

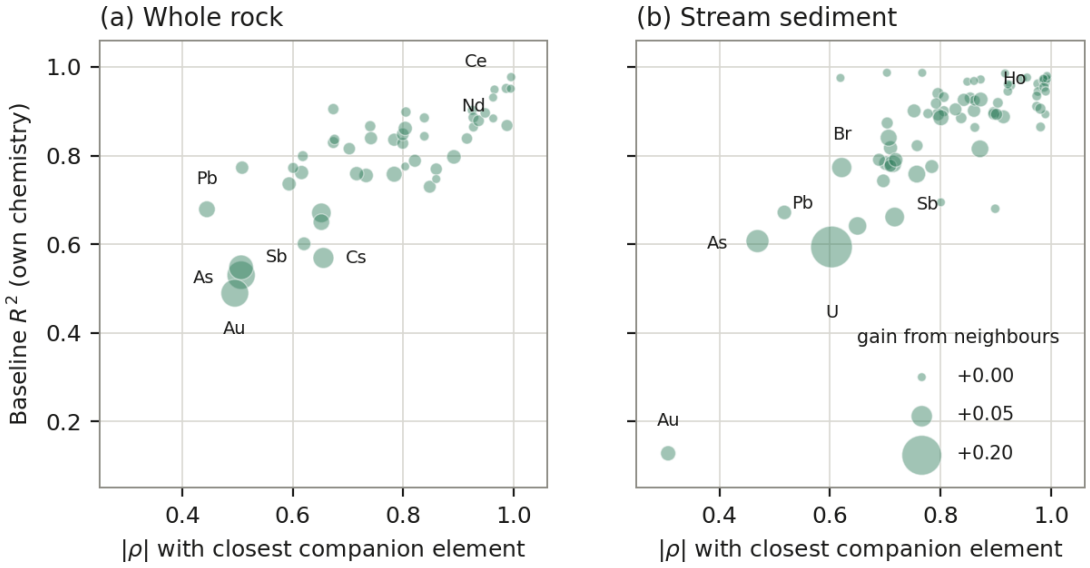


Figure 5: Baseline R^2 against companion strength, one point per target element; marker area is the gain from neighbour features. Elements with a near-perfect companion sit top right and gain nothing; chemically isolated elements sit lower left and are where the neighbourhood pays. Uranium is the extreme case on stream sediment.

Companion strength predicts the baseline on both datasets ($r = +0.77$ and $+0.76$), and predicts the neighbour gain with the opposite sign (Spearman -0.70 and -0.63). Splitting targets on whether they have any partner above $\rho = 0.8$ separates them cleanly, and the two datasets agree despite sharing neither target set nor sampling geometry: elements with such a partner average a baseline of 0.864 on whole rock and 0.920 on stream sediment and gain +0.0047 from neighbours on both, while chemically isolated elements average 0.738 and 0.782 and gain +0.0242 and +0.0253 — five times as much, on roughly half the targets in each dataset.

The mechanism this exposes is a substitution. The elements at the top right are the rare

earths, which track one another almost perfectly (Ce–Pr $\rho = 0.995$, Ho–Er $\rho = 0.993$) and are therefore predictable from a single companion in the same sample; a neighbourhood tells such an element nothing it does not already know. The elements at the lower left have no proxy at all: gold's best companion is $\rho = 0.31$ on stream sediment, uranium's 0.60, arsenic's 0.47, and none of the three has a single partner above 0.8. **Spatial context is worth something precisely where the sample's own chemistry contains no substitute for the target.**

This also unifies the two readings of Section 3.4. "Mobile element" and "hard target" are not competing explanations but the same one seen from either side: an element whose concentration is set by mineralisation or alteration rather than by bulk composition is, for that reason, one that nothing else in the sample tracks — which makes it both hard to predict and responsive to where it was found. The confound noted in Section 4.4 is not removed by this, since isolation still explains both, but it is named.

3.6 Where the spatial information comes from

Scoring each feature source alone and in combination (Figure 6) splits the question in two, and the two halves answer differently.

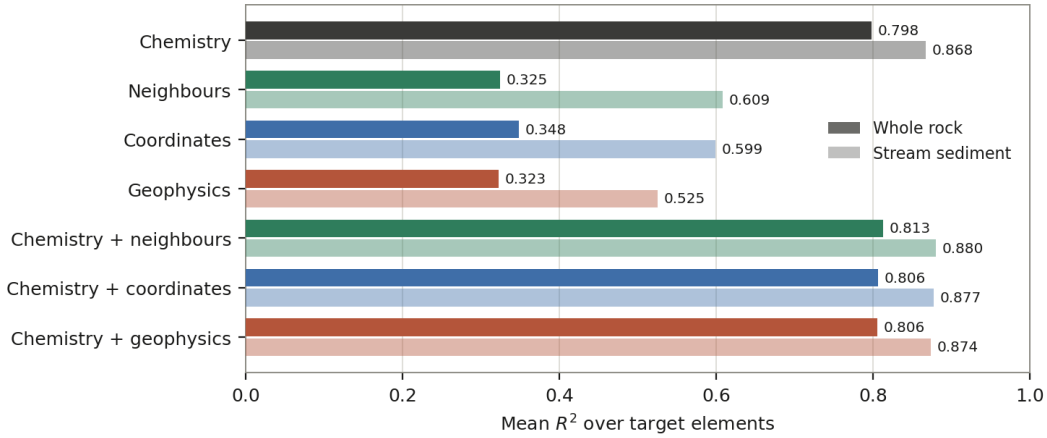


Figure 6: Mean R^2 by feature source, coverage-threshold target sets. Colour identifies the feature source, opacity the dataset. The upper block (spatial information alone) roughly doubles from whole rock to stream sediment; the lower block (the same information on top of own chemistry) barely moves.

Sampling geometry decides how much spatial information exists. Every standalone spatial source is roughly twice as informative on stream sediment as on whole rock: neighbours 0.325 against 0.609, coordinates 0.348 against 0.599, geophysics 0.323 against 0.525. This is a large and robust difference, and exactly what the dispersed-versus-clustered contrast of Figure 1 would predict.

It does not decide how much that information adds. Five times the standalone spatial signal buys the same sub-σ improvement on top of own chemistry: +0.0149 against +0.0125 for neighbours, +0.0080 against +0.0099 for coordinates, +0.0076 against +0.0063 for geophysics. The reason is redundancy — a sample's own chemistry already encodes most of what its location would tell you, because chemistry is what location predicts.

The three spatial sources are interchangeable. Geophysics, coordinates and neighbours land in one narrow band on each dataset, and each adds between +0.006 and +0.015 on top of chemistry.

In particular **the geophysical grids do not beat plain coordinates**: coordinates outscore geophysics on both datasets, and combining the two barely exceeds coordinates alone (0.6017 against 0.5994 on stream sediment). On this evidence the grids encode *where* a sample is rather than independently *what* lies underneath it.

A correlation test outside the models agrees. Comparing, for each element, its strongest correlation with a geophysical field against its strongest correlation with raw easting or northing, geophysics wins for only 27 of 64 elements (42 %) on whole rock and 26 of 81 (32 %) on stream sediment. How far a grid is a position proxy depends on the sampling footprint rather than on the grid: depth to Moho correlates $r = 0.75$ with northing across whole rock's narrow longitude band, but only -0.23 across stream sediment's full-island coverage. Section 4.4 explains why this is the reading most likely to invert under spatial cross-validation.

3.7 A learned aggregation does not do better

Figure 7 compares both graph architectures against variant C across the same seven values of k, on identical folds.

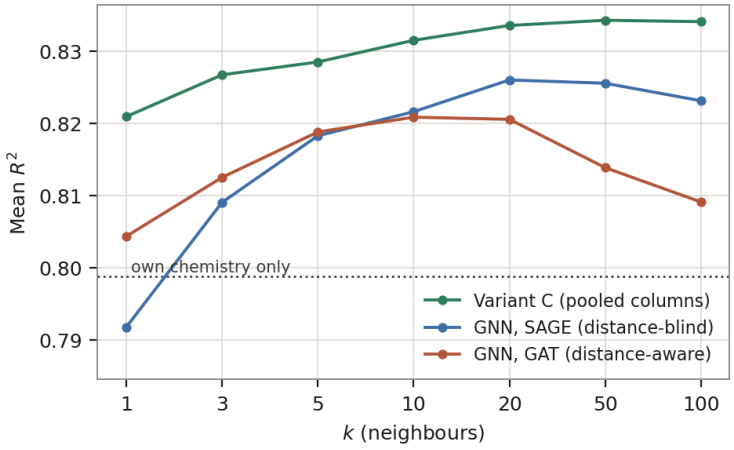


Figure 7: Mean R^2 on stream sediment for the two graph architectures and the hand-built pooled encoding, across neighbourhood size. Variant C dominates both networks at every k. Distance-aware attention (GAT) beats the distance-blind control (SAGE) only below $k \approx 5$, and degrades above $k = 20$.

The hand-built encoding wins everywhere. Variant C is above both networks at all seven values of k, with no crossover. The best network is GraphSAGE at $k = 20$ (0.8260); the best hand-built result is variant C at $k = 50$ (0.8342).

Distance-aware attention does not pay, and eventually hurts. The two architectures cross at $k \approx 5$. Below it the attention model is better, as designed — with few neighbours, distance discriminates usefully. Above it the distance-blind control wins by a margin widening monotonically to $+0.0140$ at $k = 100$, well outside the ≈ 0.002 run-to-run variation between repeats of one configuration. The attention model also degrades past $k = 20$ (0.8205 to 0.8091), tracking edge count exactly as variant D tracks column count. The same aggregation-versus-dimensionality effect appears in the learned model.

On the full target set the network matches the baseline, not the encoding. Running the settled configuration (GraphSAGE, $k = 20$) on all 71 stream-sediment targets gives 0.8676 — identical to the no-neighbour baseline, against variant C's 0.8801. Since the graph is always on,

that baseline is the wrong comparison and 0.8801 is the number to beat. One target distorts the mean: `Sum wt%` is a −0.253 outlier for the network where nothing else is worse than −0.061 (Section 4.4). Excluding it, the network is worth +0.0037 against variant C's +0.0128, recovering roughly 29 % of the available gain rather than none. On uranium it exactly ties the hand-built columns (0.814 against 0.814): it captures the one large spatial signal and misses the many small ones. Adding geophysical or coordinate channels to the node features moves the mean by +0.0004 and +0.0008 — inside noise, and a further instance of the redundancy of Section 3.6.

3.8 Robustness

Model family. The central claim is a null result, so it was checked under a second inductive bias. On whole rock all four encodings were run under both a random forest and gradient boosting, with the same conclusion under each: every gain inside one fold standard deviation, and variant D worst of the four. Under the random forest at $k = 20$ the four encodings are worth +0.0016, −0.0004, −0.0003 and −0.0101 over the no-neighbour baseline; under gradient boosting, +0.0033, +0.0033, +0.0064 and −0.0002. Stream-sediment results use gradient boosting only; variant D at $k = 20$ is 1 740 columns and `RandomForestRegressor` evaluates all features at every split, which makes the repeat prohibitive for a confirmation whole rock already provides.

Target selection. The pooled mean is far more sensitive to which elements are scored than to anything about the model, and the size of that sensitivity is worth quantifying. Scored on each dataset's ten best-covered elements — the usual convenience choice — variant C is worth 0.81σ on whole rock but 2.93σ on stream sediment, a sharp contrast inviting the reading that sampling geometry decides whether spatial context helps. Under the coverage threshold whole rock is unchanged at 0.82σ while stream sediment falls to 0.91σ, leaving the two statistically indistinguishable.

The collapse is one element. Uranium's gain on stream sediment is +0.2205, an order of magnitude above any other element's, and stream sediment's top ten happens to contain it: it supplies +0.0220 of that set's +0.0347 total, or 63 %, against +0.0031 of +0.0125 (25 %) in the 71-element mean. Excluding uranium entirely, stream sediment's gain is +0.0096. Whole rock's top ten, by contrast, is dominated by major oxides that own chemistry predicts easily, which exaggerated the contrast from the other end.

Nothing here is a numerical error — the folds are honest and the means are correct — but the apparent dataset-level effect is an artefact of a selection made before any modelling, and it survives model comparison, a k-sweep and a co-location control, because none of those checks touch target selection. This is the direct justification for reporting the stratified table of Section 3.3 rather than a pooled mean, and for choosing targets by a coverage threshold.

4 Discussion

4.1 Spatial context is redundant, not absent

RQ1 reduces to one sentence: *a sample's own chemistry already contains most of what its location could tell you.* Every route into the spatial signal — averaging neighbouring samples, three independent geophysical fields, raw projected coordinates, and a learned graph aggregation — lands in the same narrow band of improvement, all of it under one fold standard deviation. That several very different encodings converge on the same small number is itself the argument: if the limit were an encoding problem, one of them should have done materially better, where a consistent ceiling suggests the limit is informational.

This is not the same as saying spatial structure is absent. Section 3.6 shows a great deal of it: neighbours alone reach R^2 0.609 on stream sediment with no access to the sample's own chemistry

whatsoever. The signal is real, substantial, and almost entirely redundant with information the analyst already has.

Section 3.5 sharpens "redundant" into something measurable. What makes an element predictable from its companions is having a companion: the correlation between an element's strongest within-sample partner and its baseline score is +0.77, and the correlation with its neighbour gain is the same relation with the sign flipped. Spatial context is not competing with own chemistry in general — it is competing with whichever element in the same sample happens to be the target's best proxy, and it wins only where there is not one. That framing predicts the whole shape of Figure 4 without reference to space at all.

4.2 Why pooling beats enumeration

Variants C and D contain the same measurements — C averaged into m columns, D enumerated into mk — and C improves with k where D degrades in step with its column count, on both datasets and in the graph network's edge count. The interpretation is that a neighbourhood carries a small amount of signal spread thinly across many correlated measurements. Averaging concentrates it and suppresses per-sample noise at fixed model capacity; enumerating preserves the same signal but dilutes it across a feature space the learner must now search, at a cost exceeding the informational gain. The neighbour ordering that D preserves — which sample was nearest — turns out not to be worth its dimensionality, consistent with the graph network's finding that edge distance stops helping once the neighbourhood is reasonably large.

4.3 Why the graph network does not win

The network was given every structural advantage — the same folds, the same neighbourhood, a skip connection so the graph only had to add what it knew, and an architecture able to read edge distance — and still does not beat an average. Three explanations, not mutually exclusive. With only element concentrations as node features, the messages passed between neighbours carry little that a per-element neighbour mean does not already summarise, and gradient boosting is simply the stronger tabular learner at this data scale [2]. The graph is also purely geometric: a k-nearest-neighbour graph in metres encodes proximity, whereas the geologically meaningful adjacency is shared catchment, shared lithology or shared intrusive suite, none of which it can see. And the network was tuned only lightly, so some margin remains — though not, on this evidence, the +0.0125 needed to overtake an 87-column average.

4.4 Limitations

Random cross-validation. All results use random folds, and position is unusually informative when train and test interleave in space. The reading most exposed is Section 3.6's conclusion that geophysics does not beat coordinates: under spatial blocking, coordinates should degrade sharply — a model that has learned "this northing implies this uranium level" is useless when that whole band is held out — while the geophysical grids remain real measurements available at every point. That comparison could invert, and testing it is the first task of RQ2.

Two derived target columns. `Sum wt%` in the stream-sediment dataset is exactly the sum of the major oxides plus loss on ignition (median absolute difference 0.0000 % over 8 482 samples, 98.9 % within 0.5 %), and the relation runs both ways, so the major oxides are inflated by subtraction as well. Separately, about ten elements appear twice under different unit conventions (Fe as `Fe %` and `Fe2O3 wt%`, $r = 0.93$; Mn likewise, $r = 0.97$) — redundancy rather than leakage, since the pairs come from different digestions. Both raise absolute baselines and dilute measured gains. Neither affects any conclusion here, because every claim is a *difference* between two arms that

see identical features; but the > 0.95 band of Figure 4 is partly populated by these columns and should not be over-interpreted. The < 0.70 band, which carries the finding, is unaffected.

Confounded explanation of the pathfinder result. The elements that gain most are mobile and hydrothermally mobilised, they are the elements own chemistry predicts worst, and by Section 3.5 they are the elements with no close companion in the same sample. Isolation is a plausible common cause of the other two, but it is a description rather than a test: separating "spatial context helps mobile elements" from "spatial context helps elements without a chemical proxy" would require a set of elements matched on companion strength but differing in mobility, which this data does not obviously contain.

Scope. The two-model check covers whole rock only, and graph experiments stream sediment only. Five stream-sediment targets have fold standard deviations above 0.03 (Ta, Au, U, Pr, Cl), and gold is effectively unpredictable from chemistry (R^2 0.128).

5 Conclusion and future work

One pipeline, applied unchanged to two independent GEUS geochemistry datasets, answers RQ1: a held-out element can be predicted well from a sample's other elements, and spatial context adds little on top. Neighbouring chemistry, geophysical grids and raw coordinates each add less than one fold standard deviation, they are interchangeable with one another, and a graph neural network with learned distance-weighted aggregation does not beat a plain average over the same neighbourhood. Where the residual gain lands is systematic — roughly +0.05 for elements own chemistry predicts poorly, nothing for those it already predicts well, on both datasets — and measuring the chemistry directly says why: those elements are the ones with no close companion in the same sample, so spatial context pays exactly where own chemistry carries no substitute for the target. Separately, the choice of which elements to score moves the headline number more than any modelling decision examined here.

5.1 Future work: RQ2

The natural continuation is to hold out *regions* rather than samples:

1. **Spatial block folds.** Hash projected coordinates into squares of side `blockKm` and assign whole blocks to folds, written in the existing fold-file format so no downstream code changes. Blocks are preferable to clustering, because block size in kilometres is interpretable whereas clusters follow sampling density [3].

2. **Re-run the core comparison.** Baseline, variant C and the full feature-source ablation. This is the step that can change a conclusion: if geophysics overtakes coordinates under spatial hold-out, that is a substantive result and precisely the geologically motivated angle the project set out to add.

3. **A block-size sweep.** R^2 against block size gives a degradation curve — the distance over which the model generalises — which is a more useful deliverable to a geologist than any single number.

4. **Geological province hold-out.** The stricter test: a coherent terrane rather than an arbitrary square, using the published subglacial province polygons.

RQ2 will use stream sediment only. Whole rock spans 11.7° of longitude within a single region of West Greenland, so there is no meaningful "outside" to hold out, and block cross-validation would either produce blocks too small to be spatially independent or fragment the dataset. Its

role here was as the control that made the sampling-geometry comparison possible, and that role is complete.

Beyond RQ2, the direction suggested by Section 3.7 is not a better network but a better graph. The present adjacency is purely geometric; encoding shared catchment, mapped lithology or intrusive suite would give message passing something a distance matrix cannot express, which is the condition under which a graph model could be expected to overtake a well-chosen average.

Reproduction

Four notebooks are run in order: `1. Data Fetching` (preprocessing, geophysical attachment, splits), `2. Random Forest` (tabular models, neighbour encodings, k-sweep, feature-source ablation), `3. GNN` (graph construction, both architectures, k-sweep, settled configuration) and `4. Results` (cross-dataset aggregation, the correlation analyses of Section 3.5, the per-element exports, and every figure here). The first three take one dataset at a time; switching requires changing three constants. Every experiment caches to JSON under `Results/`; each figure is built from that cache by `Report/make_figures.py`, which the last section of `4. Results` imports and renders inline. Seeds are fixed at 42; only the graph experiments need a GPU.

Declaration on the use of AI tools

This project used Claude (Anthropic) via the Claude Code environment [4] as a programming assistant for the pipeline, experiments and figures, and in drafting this report from the experimental outputs. All reported numbers are produced by running the included notebooks against the committed data. I have reviewed and verified the content and take full responsibility for it.

References

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[3] David R. Roberts et al. "Cross-validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure". In: *Ecography* 40.8 (2017), pp. 913–929. DOI: 10.1111/ecog.02881.

[4] Anthropic. *Claude and Claude Code*. https://claude.com/claude-code. Accessed 2026. 2026.